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ABSTRACT

Flame stabilization is critically challenging in the advancement of scramjet engines, as it directly affects their operational efficiency and safety. Consequently, accurately predicting the future combustion state of the scramjet flame is essential for implementing proactive measures to mitigate potential safety risks while enhancing fuel efficiency. In this paper, we report a Simpler yet Better Video Prediction (SimVP) model based on kerosene-fueled scramjet combustion images for spatiotemporal prediction. By establishing a temporal relationship between historical image sequences and future ones, the upcoming combustion state of the flame can be generated. In this work, the predictive capability of the proposed method was tested by varying the number of predicted frames. This study compared typical spatial-temporal neural networks, including Recurrent Neural Network (RNN) based models, Temporal Attention Unit model, and the SimVP network, with the last one demonstrating the best feature capture capability and hence prediction performance. It was able to predict seven future flame frames based on an input sequence of eight frames, while maintaining mean absolute error of 25.6896×10^{-3} , mean square error (MSE) of 7.2821×10^{-3} , peak signal-to-noise ratio of 23.0342, and structure similarity index measure of 0.9990, reducing MSE by 8.04% and 89.31% compared with ConvLSTM and PredRNN, respectively. The SimVP method reported in this work, due to its superior accuracy and novel architecture, holds promise as a potential approach for spatiotemporal prediction of supersonic turbulent combustion.

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I. INTRODUCTION

As the most promising candidates for future air-breathing systems in hypersonic flight, scramjet engines have received increasing attention in recent years.^{1–7} The study of hypersonic flame stabilization plays a crucial role in the safe and efficient operation of scramjet engines. Different local equivalence ratios lead to changes in the combustion mode, which in turn trigger pressure oscillations in the scramjet engine.^{2,8,9} The pressure oscillations coupled with heat release rate fluctuations give rise to combustion instability, which in turn leads to high levels of noise and vibration^{10–12} and may eventually pose risks to safe operation, result in structural damage and catastrophic failure, and cause significant economic losses.^{12,13} To suppress and control combustion instability, it is necessary to develop a method that can monitor and predict turbulent combustion conditions. Traditionally,

the combustion process can be interfered with by controlling dynamic pressure with passive^{14–20} and active methods.^{21–26} However, the measurement of the dynamic signal is affected by engine vibrations, resulting in a low signal-to-noise ratio. High-speed cameras and advanced optical imaging techniques have become competitive solutions due to their non-intrusive, real-time, *in situ* measurement capabilities, high spatiotemporal resolution, and applicability in high-temperature and high-pressure environments.²⁷ In recent years, the study of supersonic combustion phenomena has highlighted the significant role of flame structures in revealing the mechanisms of flame stabilization and improving the design of scramjet engines.²⁸ Optical measurement techniques, owing to their unique advantages in visualizing flame structures, have been applied not only in steady-state combustion but also in the diagnosis of complex transient combustion scenarios.^{28–31}

Images captured by high-speed cameras can be analyzed to acquire combustion mode by Proper Orthogonal Decomposition (POD) as was reported by Nakaya *et al.*³² Such algorithms detect unstable combustion by allowing for transforming spatial information of flames into the frequency domain, but they cannot be applied in monitoring and alarming systems because of the lack of ability to predict future combustion status. Also, the POD approach requires prior knowledge and lacks generalization capability, which results in inaccuracies and inefficiencies when processing a large amount of combustion data. Machine learning architectures, particularly deep learning frameworks, have made explosive progress due to the advancement of mathematical optimization and computational hardware, positioning them as efficient methods for processing a large amount of higher-dimensional data. Owing to its proven effectiveness in image processing tasks, Convolutional Neural Network (CNN)^{33,34} has been a promising method for combustion image analysis. Choi *et al.*³⁵ developed a 2D CNN model to enhance the accuracy of combustion state prediction. Ghosal *et al.*³⁶ formulated combustion instability detection as binary classification using 3D CNN architecture. Sarkar *et al.*³⁷ proposed a framework that combines CNN and Symbolic time series analysis for the early detection of instability. However, these works are not preferable to understand the temporal transitions of turbulent flame structures, and thus they cannot generate predicted flame images based on a series of previous combustion states. To address this issue, a temporal sequence prediction model that takes an image sequence (recording the dynamics of turbulent combustion) as the input and a future image sequence (predicting the future combustion states) as the output needs to be proposed. Some researchers have attempted to construct such models using Recurrent Neural Networks (RNN)³³ with CNN and its variant models such as long short-term memory (LSTM)³⁸ and gated recurrent unit (GRU).³⁹ Patraucean *et al.*⁴⁰ combined optical flow with RNN to establish a temporal dynamics model. However, this model is difficult to apply due to the high computational costs. Zhang *et al.*⁴¹ proposed a Super-SloMo method, and Guo *et al.*³⁰ presented the Conv-LSTM network to form the computational model for turbulent combustion interpretation. However, these two approaches can only perform interpolation of a short image sequence and are unable to achieve long-term prediction.

Over the past few years, numerous advanced methods have emerged in the field of deep video prediction, with one of the state-of-the-art solutions utilizing stacked RNN modules as autoencoders within a recurrent pyramid structure. Shi *et al.*⁴² reported a ConvLSTM network that extends fully connected LSTMs and its convolutional computing structure, which demonstrated the capability of capturing spatiotemporal correlations. Wang *et al.*⁴³ proposed PredRNN to extract and memorize spatiotemporal characters simultaneously. Although these methods with high flexibility have excellent performance on various prediction tasks, the need for sharing states leads to high computational and memory costs. In addition to RNN-based methods, some scholars have explored the use of Vision Transformer (ViT)⁴⁴ to model latent video dynamics, sparking a wave of interest and further research in this direction. Neimark *et al.*⁴⁵ applied a sliding window attention mechanism and 2D spatial feature to capture temporal and spatial features. This network is able to learn temporal information from image sequences, but it is not designed for prediction tasks. Latent video transformer presented by Rakhimov *et al.*⁴⁶ and the scaled autoregressive video models reported by

Weissenborn *et al.*⁴⁷ both applied an attention mechanism to deal with the problem of video prediction under specific circumstances. However, their models lack generalizability across different datasets.⁴⁸ Recently, CNN-based recurrent frameworks have been developed to address the problem of high computational cost and generalization. Additionally, Chiu *et al.*⁴⁹ used CNN and teacher-student distillation to segment the future image sequences. This work has achieved video prediction with a simple framework and low computational cost. However, it requires complex modules and training strategies, which pose a challenge for implementation in general situations. Gao *et al.*⁵⁰ reported a SimVP model with flexible predictive length and excellent performance across different scenarios. Although SimVP builds its encoder and decoder on CNN, which increases the simplicity and flexibility, it still requires a relatively complex attachment (Inception-UNet) to achieve substantial performance. Tan *et al.*⁵¹ replaced SimVP's Inception module with a Temporal Attention Unit to formulate a simpler structure.

This paper presents a computational imaging model based on the SimVP network, which utilizes a combination of static and dynamic attention mechanisms to extract temporal information of image sequences, while employing CNNs as both the encoder and decoder. Compared to recurrent-based networks, this model is simpler, smaller in size, and capable of achieving high-accuracy long-term predictions of supersonic flame structures. Experiments were carried out in a kerosene-fueled scramjet combustor using a multi-channel gliding arc (MCGA) plasma. The model's hyperparameters have been optimized to achieve an efficient model structure, and its performance has been compared with RNN-based models and TAU. Additionally, by adjusting the ratio of input to output frames, the predictive limits of the model were evaluated using three different metrics, while the prediction of flame propagation was assessed through the intersection of the union. At this limit ratio, statistical data on the accuracy of flame propagation prediction were comprehensively presented. Furthermore, the intensity and intermittency profiles of the flame images demonstrated the model's strong predictive performance. Finally, the detailed tracking of flame propagation and the robustness of the model were analyzed in this study.

II. EXPERIMENTAL DATA

In the present work, the scramjet combustion data were obtained from experiments conducted in a scramjet combustor with multi-channel gliding arc (MCGA) igniters. A more detailed experimental setup can be found in previous papers.^{3,28,52–56} Figure 1 illustrates a schematic of the scramjet combustor, which includes three sections: a cross-sectional isolator, a supersonic model combustor, and an expansion section. The cross-sectional isolator is the region where the supersonic vitiated air flow mixes with kerosene injected by the injectors. To inject high-speed kerosene, a Laval nozzle designed for $Ma = 2.62$, with $T_t = 1650$ K and $P_t = 2.6$ MPa, was used. In the isolator, upstream of the cavity's leading edge, two 1 mm diameter vertical fuel injectors were positioned. The entrance isolator, designed with dimensions of 50 mm in width and 40 mm in height, leads to a cavity that measures 64 mm in length, 50 mm in width, and 16 mm in height, with a 45° back ramp. Two MCGA igniters, aligned with the lower edge of the back ramp, were used in the supersonic model combustor. A quartz glass window mounted on the sidewall provides optical access of 60×200 mm² to facilitate high-speed camera acquisition of scramjet combustion images. Each igniter, featuring a six-channel gliding arc,

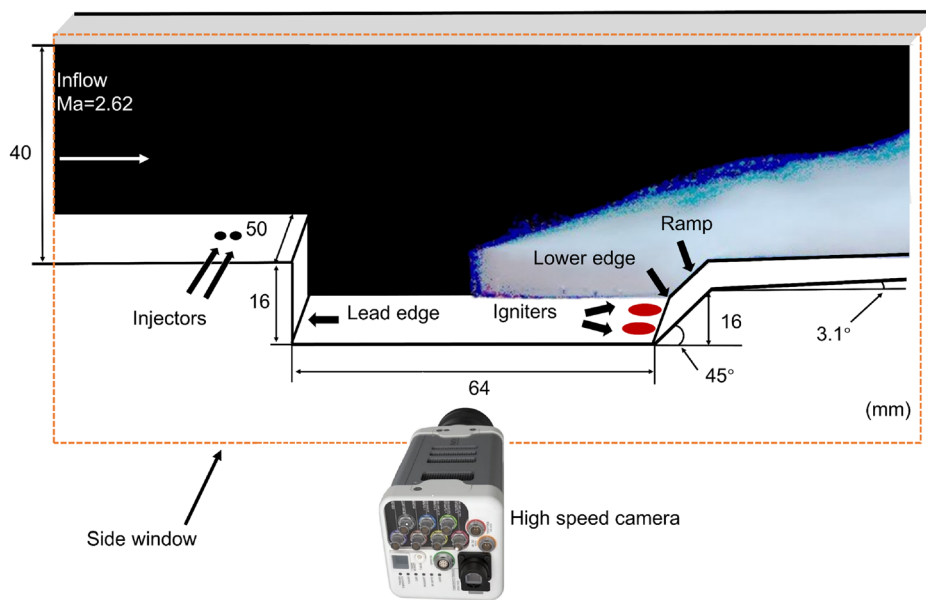


FIG. 1. Scramjet cavity with the MCGA plasma.

was utilized to enhance combustion. The breakdown distance for the gliding arc channels was 3 mm, measured between the electrodes and the iron wall. The expansion section applied an expansion angle of 3.1° to enhance the primary vortex circulation and reduce residence time within the cavity.

Through the side window, a high-speed camera (SA-X2) with a high-speed camera intensifier (UVi model 1850) and an OH narrow-band filter (310 ± 10 nm) recorded sequential OH^{*} chemiluminescence images from the side view of the combustor. The SA-X2 camera operated at a frame rate of 20 kHz with an exposure time of 48 μ s, triggered by an oscilloscope. The ignitor with six tungsten-made high-voltage (HV) electrodes, a filled ceramic, and a stainless-steel case generated MCGA plasma using an HV alternating current power supplier (CG10000FG, Suman Technology) and plasma generator (CTP-2000 K, developed by Suman Technology). Discharge waveforms were measured with a high-voltage probe (P6015A) and a current probe (TCP0030A). Additionally, the emission spectra were recorded by a spectrometer (Ocean Optics) to characterize the radicals produced by the MCGA discharge in the airflow.

In this work, all images underwent gamma correction to enhance the distinction between reactive and non-reactive regions of the flame (Fig. 2). During this process, the pixel intensity is first linearly mapped from (0, 255) to (0, 1), followed by a gamma power transformation, and then the transformed pixel values are remapped back to (0, 255). The mathematical expression for the gamma transformation is as follows:

$$s = f(r) = cr^\gamma, \quad r \in (0, 1). \quad (1)$$

Here, c is a constant, set to 1 in this work; r represents the pixel values after linear mapping; γ is the gamma value, which is set to 0.1; and s denotes the pixel values before being remapped.

The kerosene-fueled scramjet combustion experiments were conducted under three conditions listed in Table I, namely with fixed $Ma = 2.62$, $T_1 = 1650$ K, and $P_1 = 2.6$ MPa and variant Kerosene mass

flow rate m_k and equivalence ratio (ER). As shown in Fig. 2, the recording area is -60 to 150 mm in length and -20 to 80 mm in height. The image size was 896×368 in pixel,² which is equivalent to a spatial resolution of 0.27 mm/pixel. There were 9000 snapshots of flames acquired in total. For each flame condition, a temporal sliding window of 15 was applied to generate image sequences, resulting in a dataset comprising 8961 sequences of uniform length.

III. METHODOLOGY

A. Preliminaries

The spatiotemporal scramjet flame predictive model requires a combustion image sequence that contains input series $\chi^{t,T} = \{x^i\}_{t-T+1}^t$ and target series $\gamma^{t+1,T'} = \{x^i\}_{t+1}^{t+1+T'}$, here $x_i \in \mathbb{R}^{C \times H \times W}$ is usually an image with channels C , height H , and width W . It should be noted that the model processes sequential data in batches, which helps reduce the time required for training. This is the reason that the input data has an additional dimension B (batch size), and in this study, this dimension is fixed at 16 to account for computational cost. Input images include the sequence at time t along with the past T frames, while target images include the sequence with the subsequent T' frames. According to the description above, input images have the dimension of $\mathbb{R}^{T \times C \times H \times W}$ and target images have the dimension of $\mathbb{R}^{T' \times C \times H \times W}$. The model proposed aims to optimize the internal parameters Θ to learn a mapping $F_\Theta : \chi^{t,T} \rightarrow \gamma^{t+1,T'}$ by exploring both spatial and temporal dependencies. To minimize the difference between the prediction and ground truth, the mapping (neural network model) needs to be trained to get the optimal parameters. The optimal parameters Θ^* are

$$\Theta^* = \arg \min_{\Theta} L(F_\Theta(\chi^{t,T}), \gamma^{t+1,T'}), \quad (2)$$

where L is a loss function that evaluates such differences.

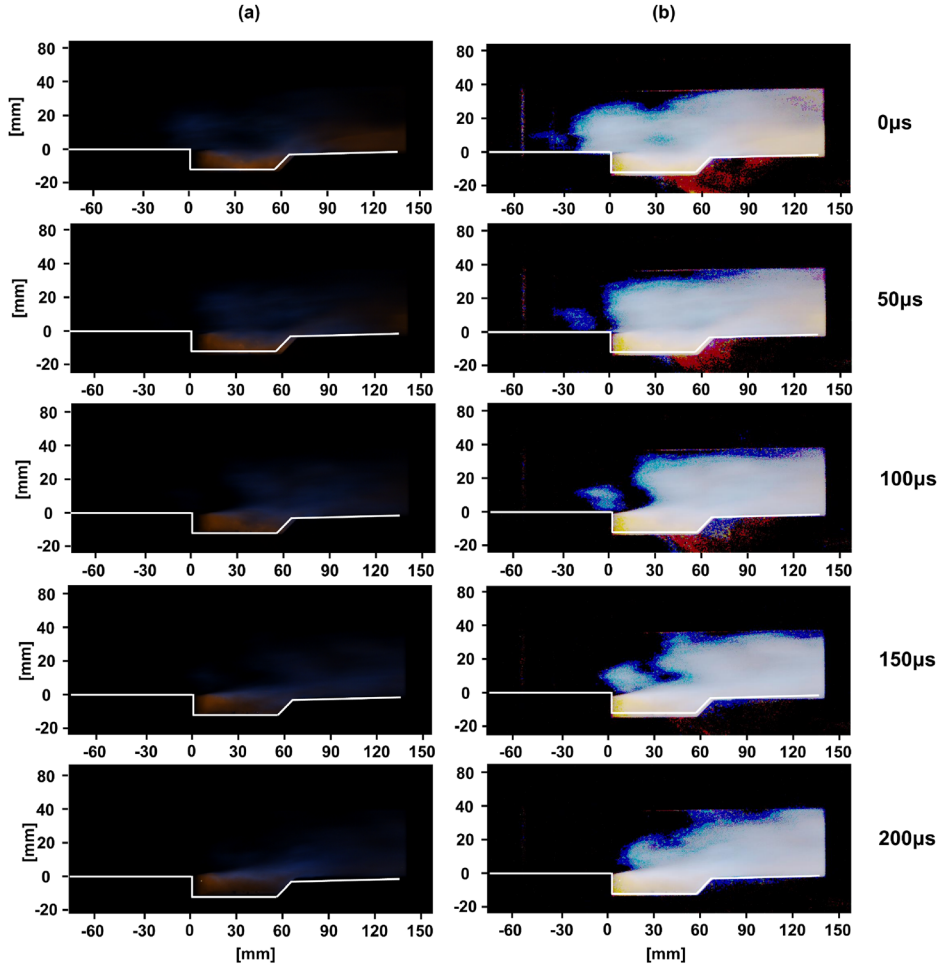


FIG. 2. (a) Representative raw scramjet combustion images from the optical experiment and (b) The corresponding images under gamma correction.

B. Model description

As shown in Fig. 3, the SimVP model consists a spatial encoder, translator, and a spatial decoder, which are all built on CNN. In the spatial encoder and decoder, the sequential input data $B \times T \times C \times H \times W$ is reshaped as $(B \times T) \times C \times H \times W$ so that only spatial correlations are taken into account. In the temporal module, we reshape the

feature $B \times T \times C \times H \times W$ as $B \times (T \times C) \times H \times W$ so that frames are arranged in order on the channel dimension. Spatial features of combustion image sequences can be extracted by an encoder that stacks N_s ConvNormReLU blocks (Conv2d+LayerNorm+LeakyReLU). The hidden feature can be expressed by the following formula:

$$z_i = \sigma(\text{LayerNorm}(\text{Conv2d}(z_{i-1}))), \quad 1 \leq i \leq N_s, \quad (3)$$

where σ represents the activation function, which is LeakyReLU, and the input $z_{i-1} \in \mathbb{R}^{(B \times T) \times C \times H \times W}$ and output $z_i \in \mathbb{R}^{(B \times T) \times C_s \times H \times W'}$. The decoder consists N_s unConvNormReLU blocks (ConvTranspose2d+GroupNorm+LeakyReLU) so that the frames can be reconstructed. The formula below shows the hidden feature in the decoder

$$z_j = \sigma(\text{GroupNorm}(\text{unConv2d}(z_{j-1}))), \quad N_s + N_t \leq j \leq 2N_s + N_t, \quad (4)$$

where N_t represents the number of Inception modules and the output $z_j \in \mathbb{R}^{(B \times T') \times C \times H' \times W'}$ and input $z_{j-1} \in \mathbb{R}^{(B \times T') \times C_s \times H' \times W'}$. To learn temporal features, the translator employs N_t Inception modules, and each of the modules consists of a bottleneck Conv2d with 1×1 kernel

TABLE I. Experimental conditions for the flame datasets.

Flame condition ^a	Number of frames	Number of sequences
mk = 29.2, ER = 0.59	3000	2390 (For train) 597 (For test)
mk = 21.7, ER = 0.43	3000	2390 (For train) 597 (For test)
mk = 12.2, ER = 0.24	3000	2390 (For train) 597 (For test)
		Total 8961

^aThe listed condition parameters are shared across the three operating conditions, specifically: Ma = 2.62, Tt = 1650 K, and Pt = 2.6 MPa.

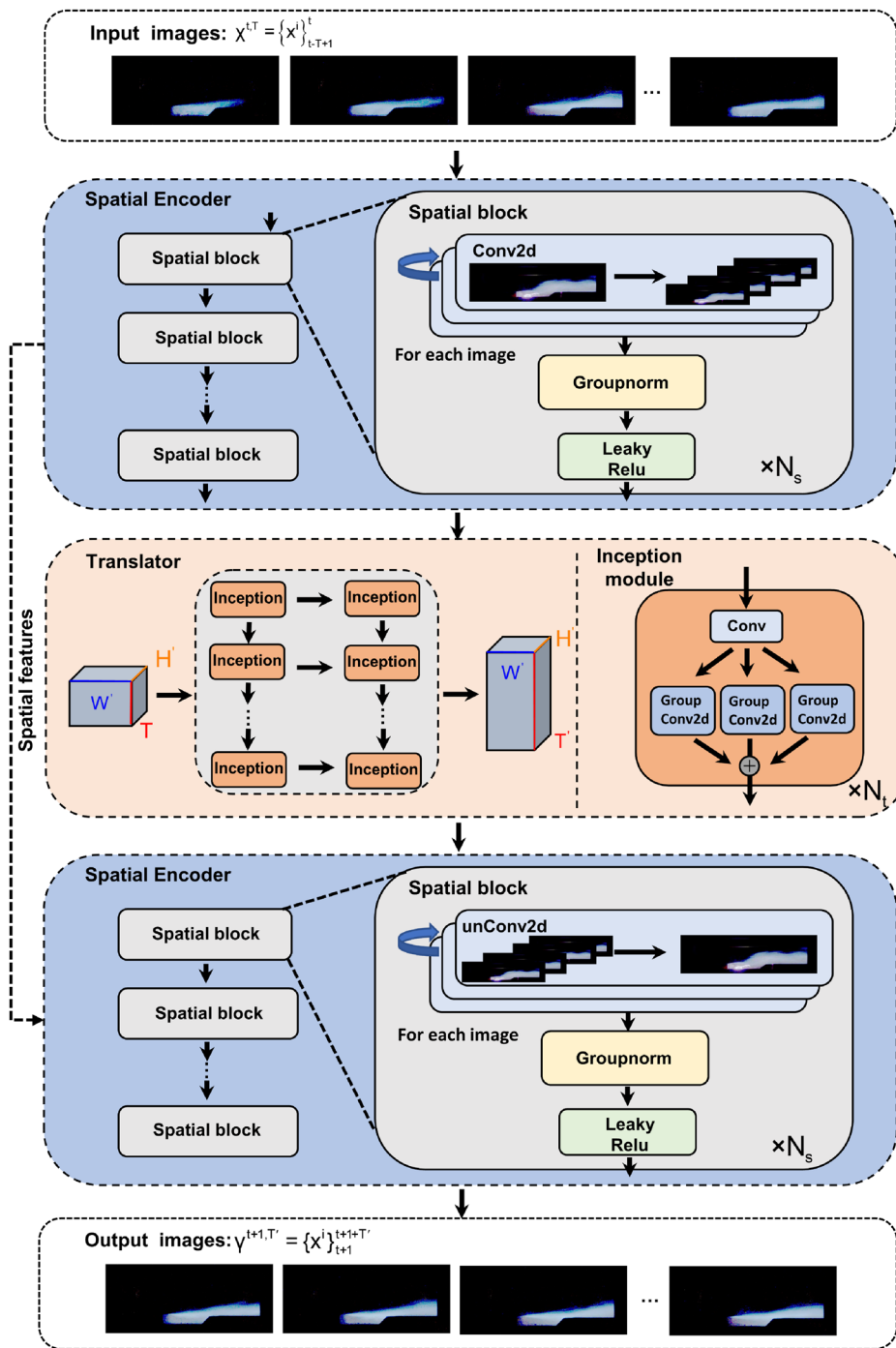


FIG. 3. The encoding-translating-decoding architecture used in the SimVP spatiotemporal model.

followed by parallel GroupConv2d operators. In Inception modules, the hidden feature can be expressed as follows:

$$z_k = \text{Inception}(z_{k-1}), \quad N_s \leq k \leq N_s + N_t, \quad (5)$$

where the output is $z_k \in \mathbb{R}^{B \times (T' \times C_s) \times H' \times W'}$ and input is $z_{k-1} \in \mathbb{R}^{B \times (T \times C_s) \times H \times W}$.

The scramjet combustion image sequence is split into two sets: a training set containing 80% of the data and a test set containing the last 20%. The training set is utilized for optimizing the internal parameters of the model through backpropagation,⁵⁷ while the test set is used for calculating and evaluating the performance of the trained model.

In the training process, we utilized the Adam⁵⁸ optimizer with a maximum learning rate set to 10^{-4} and adopted a one-cycle⁵⁹ schedule to minimize the mean square error (MSE) loss. MSE of predicted f and its ground truth g can be computed by the following equation:

$$\text{MSE} = \frac{1}{m \times n} \sum_{i=1}^m \sum_{j=1}^n (f_{ij} - g_{ij})^2, \quad (6)$$

where f_{ij} and g_{ij} represent the pixel intensity of image f and g at row i and column j , respectively; m and n represent the number of rows and columns, respectively.

All models were implemented in Python using the PyTorch Lightning framework. The computation was completed on two NVIDIA Tesla V100 GPUs. The dataset comprises 9000 images captured by the previously mentioned optical system. These images represent three distinct experimental conditions, and for each condition, image sequences were generated using a temporal sliding window of 15 in length. The models evaluated in this study were trained on 7170 scramjet combustion sequences and tested on 1791 sequences, each consisting of 15 images. By varying the ratio of input images to predicted images, we investigated the predictive capability of the model. Multiple tests were conducted, ranging from an input of 14 images producing a single output image to an input of 8 images yielding seven output images.

C. Quantitative evaluation of model accuracy

To quantitatively evaluate the accuracy of the model, following Refs. 60 and 61, we employed several metrics to quantify the difference between predicted image P and experimental supervision T . In addition to MSE mentioned in Sec. V, four indices, i.e., mean absolute error (MAE), intersection over union (IoU), peak signal-to-noise ratio (PSNR),⁶² and structure similarity index (SSIM),⁶³ were adopted in this research. MAE measures the average magnitude of errors between predicted and true pixel values. The computation of MAE is as follows:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |P_i - T_i|, \quad (7)$$

where P_i and T_i represented the prediction and measurement results of the i th element, respectively. The N was the total number of predicted elements. As shown in Eq. (8), IoU is used to quantify the prediction accuracy of signal occurrence by evaluating the overlap between the predicted and ground truth regions. In Fig. 5, the red region represents the difference between the predicted image and experimental supervision, and the length of a sequence and predicted frames are indicated as subscripts. For example, P_{14-1} means using a length of 14 image sequences to predict 1 image. Gamma correction with a value of 0.1 was performed to enhance the signal-to-noise ratio of the original images.

$$\text{IoU} = \frac{\hat{P} \cap \hat{T}}{\hat{P} \cup \hat{T}}. \quad (8)$$

As a widely used metric, PSNR stands for the ratio between the maximum possible power of a signal and the power of corrupting noise that affects the fidelity of its representation, indicating reconstruction quality for images and video subjects. The PSNR ranges from 0 to ∞ , with higher PSNR representing higher prediction accuracy. A PSNR of 20 dB and above⁶³ suggests that the model has well-reconstructed the

images with high similarity. To compute PSNR, information of pixel difference between two images (MSE) is needed, and the equation is as follows:

$$\text{PSNR} = 10 \times \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right). \quad (9)$$

SSIM, which is shown in Eq. (10), is used for measuring the similarity between two images by comparing their structural information, luminance, and contrast.

$$\text{SSIM}(P, T) = l(P, T)c(P, T)s(P, T), \quad (10)$$

where the relevance of luminance l , contrast c , and structure s are further defined as follows:

$$\begin{cases} l(P, T) = \frac{2\mu_P\mu_T + C_1}{\mu_P^2 + \mu_T^2 + C_1}, \\ c(P, T) = \frac{2\sigma_P\sigma_T + C_2}{\sigma_P^2 + \sigma_T^2 + C_2}, \\ s(P, T) = \frac{2\sigma_{PT} + C_3}{\sigma_P\sigma_T + C_3}, \end{cases} \quad (11)$$

where μ_P and μ_T represent the mean value of pixel intensity for image P and T ; σ_P , σ_T represent the standard deviation of image P and T ; σ_{PT} represents the covariance of image P and T . C_1 , C_2 , and C_3 are constants to avoid a zero denominator. The constants are determined as follows: $C_1 = (K_1 \times L)^2$, $C_2 = (K_2 \times L)^2$, $C_3 = C_2/2$; $K_1 = 0.01$, $K_2 = 0.03$, $L = 255$.

D. Loss function

The model adapts a loss function that combines MSE loss and a difference divergence regulation term that can help the model learn the differences between consecutive frames. To formulate the loss function, the forward difference between predicted frames $\hat{\gamma} = F_{\Theta}(\chi)$ and ground truth frames γ needs to be calculated

$$\Delta\hat{\gamma}_i = \hat{\gamma}_{i+1} - \hat{\gamma}_i, \quad (12)$$

$$\Delta\gamma_i = \gamma_{i+1} - \gamma_{i+1}. \quad (13)$$

The difference then transferred to possibilities by utilizing softmax function

$$\text{softmax}(\Delta\hat{\gamma})_{i,j,k,l} = \frac{\exp(\Delta\hat{\gamma}_{i,j,k,l}/\tau)}{\sum_{j'=1}^C \sum_{k'=1}^H \sum_{l'=1}^W \exp(\Delta\hat{\gamma}_{i',j',k',l'}/\tau)}, \quad (14)$$

$$\text{softmax}(\Delta\gamma)_{i,j,k,l} = \frac{\exp(\Delta\gamma_{i,j,k,l}/\tau)}{\sum_{j'=1}^C \sum_{k'=1}^H \sum_{l'=1}^W \exp(\Delta\gamma_{i',j',k',l'}/\tau)}, \quad (15)$$

where τ represents the temperature parameter which is empirically set as 0.1 to sharpen the probability distribution. The high difference frames are penalized by the competition mechanism.^{64–66} The regularization term is defined as the Kullback–Leibler divergence between these two probabilities

$$L_{\text{reg}}(\hat{\gamma}, \gamma) = \sum_{i=1}^{T'-1} \text{softmax}(\Delta\hat{\gamma}_i) \log \frac{\text{softmax}(\Delta\hat{\gamma}_i)}{\text{softmax}(\Delta\gamma_i)}. \quad (16)$$

The overall loss function is described as the combination of MSE loss and the regularization term weighted by a constant α

$$L = \sum_{i=1}^{T'} \|\hat{y} - y\|^2 + \alpha L_{reg}(\hat{y}, y). \quad (17)$$

IV. RESULTS AND DISCUSSION

A. Architecture optimization and parametric study

Prediction models with different architectures exhibit varying image processing capabilities, and therefore, it is essential to compare their performance. The SimVP model (employed in the current work) was compared with some other mainstream deep learning architectures like Conv-LSTM, Pred-RNN, and TAU. The comparison was performed under the case of P₈₋₇. To directly compare their performances, for the Conv-LSTM and PredRNN, the convolutional kernel size was fixed at 5×5 , with four hidden layers and 128, 128, 128, 128 hidden states, respectively. For the SimVP and TAU models, the four hyper-parameters of the network structure—spatial block number (N_s), temporal block (N_t), hidden spatial state (h_s), and hidden temporal state (h_t)—were fixed at (2, 8, 32, 256). These models are trained on 7170 image sequences and tested on 1791 image sequences, with the length of each sequence being 15. A comparative test of training time was conducted with the batch size set to 16. The images were resized from [896, 368] to [224, 92] to consider time and computational cost.

Table II shows that the PredRNN network results in relatively large MSE and MAE, as well as a low PSNR, which is mainly because the PredRNN model has a large number of parameters and an overly complex structure, which increases the difficulty of convergence. Within the training time limit we set (three days), the training loss did not converge. As an RNN-based network architecture, ConvLSTM has a reduced number of model parameters and lower complexity compared to PredRNN. This enables it to converge within a shorter time. However, its overall performance still falls behind that of network architectures without recurrent modules, namely SimVP and TAU. As CNN encoder can extract decent and compressed spatial features, traditional RNN-based architecture is less likely to make accurate and efficient predictions without sufficient spatial information. In addition, the RNN-based network architecture, due to its complex structure (memory gates, memory decoupling operations), requires more computation compared to directly performing convolution operations. This results in increased training time and poses a challenge to training

TABLE II. Performance of the SimVP Model in Predicting 7 Frames under Different Neural Network Architectures.

	ConvLSTM	PredRNN ^a	SimVP	TAU
Parameters (M)	15.5	23.7	10.9	15.6
MSE ($\times 10^{-3}$)	7.919	68.149	7.332	7.383
MAE ($\times 10^{-3}$)	26.920	102.432	25.904	26.001
PSNR	22.601	12.561	22.980	22.868
SSIM ($\times 10^{-2}$)	99.889	98.916	99.902	99.895
Training time	56H 30MIN	More than 3D	16H 47MIN	16H 28MIN

^aThe model failed to converge within the given time of training.

costs. SimVP with inception modules outperforms TAU because its translator incorporates the Group Convolution operation, which allows it to accurately capture the dynamic features of images, whereas the motion of the scramjet flame presents strong temporal correlations with vibrant dynamic features. In Fig. 4, the statistical results of the three tested networks are illustrated. Since PredRNN failed to converge, it was excluded from the statistical comparison. Figure shows that for SimVP, most tested videos have an MSE below 0.25, an MAE below 0.01, and a PSNR above 20. Among the tested data, there are almost no cases where the MSE exceeds 0.05, the MAE exceeds 0.015, or the PSNR falls below 17.5. Compared to the TAU model, which has a similar average performance, SimVP demonstrates a particularly noticeable advantage in PSNR across 1791 test samples. According to previous research,⁶³ a PSNR value greater than 22.5 indicates relatively high prediction accuracy. In our testing, 58.85% of the results produced by the SimVP model exceeded this threshold, while that for TAU is 54.02%, demonstrating its strong performance in terms of predictive accuracy. As shown in Table III, SimVP exhibits the smallest standard deviation across all evaluation metrics, combined with the results in Fig. 4, which show that its test outcomes are concentrated around low MAE and MSE values and high PSNR values, indicating that SimVP outperforms both TAU and ConvLSTM in terms of consistent and reliable performance.

The performance of the SimVP model at different configurations was also tested and summarized in Table IV. The parentheses in the model represent the four hyperparameters of SimVP, listed in order as spatial block number (N_s), temporal block number (N_t), hidden spatial state (h_s), and hidden temporal state (h_t). The hyper-parameter tuning strategy involves using several hyper-parameters that perform well on a common dataset as Ref. 48, and then making small adjustments to the four hyper-parameters for testing. In the testing based on 1791 scramjet image sequences, the model size is also taken into consideration, in addition to model performance. This is because larger models require more computational resources at the same batch size, thereby affecting training time and cost.

In the model column of Table IV, the bolded hyperparameter settings are those that performed well on the public dataset. As can be seen from the results in Table IV, increasing the model's parameters does not lead to better performance for the scramjet combustion prediction. This is because an excessive number of model parameters may cause the model to learn features unrelated to the flame structure, such as noise in the images, which in turn leads to overfitting during the training process. In the tested model structures, a comparison of four hyperparameters on prediction performance shows that more hidden spatial and temporal states significantly increase the model size but do not lead to improvements in model prediction performance. This suggests that in the scramjet combustion prediction task, spatial and temporal information can be effectively represented with fewer features. Increasing only the hidden spatial state leads to a significant decline in model performance, which is consistent with the previous analysis. That is, an excessive number of spatial parameters causes the model to overfit the spatial features. Increasing the spatial block number and temporal block number has little impact on model performance but leads to an increase in model size. Therefore, we choose to set these four parameters to relatively small values.

In short, the network in the present work was trained by 50 epochs with 2 spatial blocks, 6 temporal blocks, 32 hidden spatial states, and

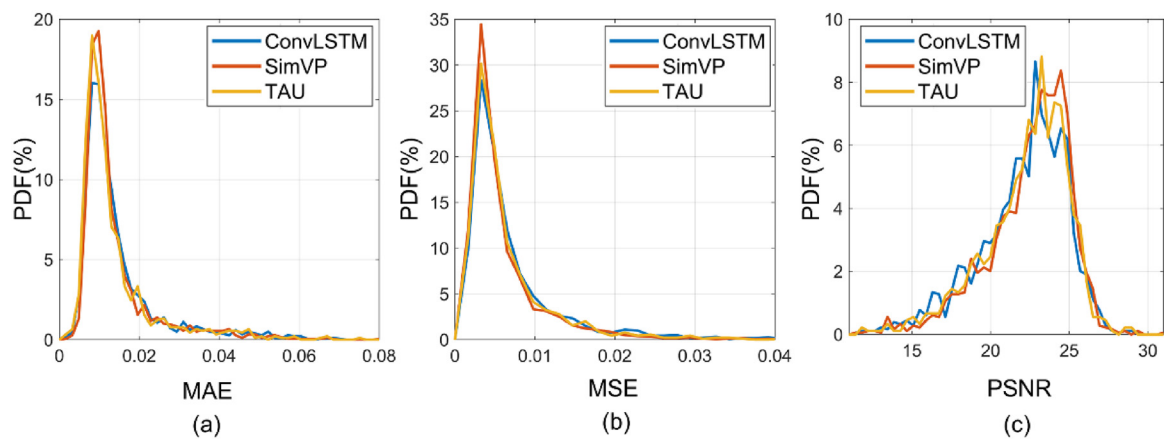


FIG. 4. Statistical performance of four tested models based on 1791 test videos.

TABLE III. Standard deviation of the three tested models across different evaluation metrics.

	SimVP	TAU	ConvLSTM
MSE	0.006	0.007	0.007
MAE	0.013	0.014	0.015
PSNR	2.587	2.640	2.746

256 hidden temporal states. Both the input-to-state and state-to-state convolutional kernel sizes were 5×5 , and the total length of scramjet combustion sequences fed to the network was 15. We evaluated the model’s real-time processing capability by measuring its inference time. The experiments were conducted on a single Tesla V100 GPU, where we performed a total of 300 trials for 7-frame prediction. To account for hardware warm-up effects, the first 50 trials were excluded. The average inference time of the remaining 250 trials was 3.731 ms, which is within the acceptable range for practical industrial applications.

B. Transient performance of the SimVP model

The performance of the model under different lengths of temporal prediction was presented in Fig. 5. It can be seen that images of

scramjet combustion flames can be generated using corresponding images from the initial time steps of the same sequence by SimVP. The performance of the studied cases is evaluated using the IoU (Intersection over Union) between the predicted and experimental images, with discrepancies between the predicted regions and experimental data visually highlighted in red. It highlights which parts of the scramjet flame structure are accurately predicted and which parts contain erroneous predictions. By comparing the experimental data (first row) with prediction results, it is evident that a larger red area indicates a higher number of prediction errors, which corresponds to a lower IoU. The case of P_{14-1} particularly demonstrates the feasibility of the method proposed in this study, as the fluidity and profile of the flames in the generated images closely resemble the experimental data. Under information-constrained conditions, predicting the image at the final time step poses the greatest challenge for the model. Therefore, comparing the discrepancy between the last predicted frame and the ground truth across different prediction horizons is of particular importance. Figure 6 illustrates the experimental images of this case, along with the predicted images, showing the column-wise averaged pixel intensity profiles. It can also be observed in Fig. 6 that the experimental values align well with the predicted values in terms of the averaged pixel intensity profile, when the prediction sequence is relatively short. While comparing the different cases studied, the prediction

TABLE IV. Performance of SimVP model in predicting 7 frames under different structures.

Model	Param	MAE($\times 10^{-3}$)	MSE($\times 10^{-3}$)	PSNR	SSIM
SimVP (2, 6, 32, 256)	8.2 M	25.6896	7.2821	23.0342	0.9990
SimVP (4, 6, 32, 256)	8.3 M	28.7546	7.8096	22.6669	0.9989
SimVP (2, 8, 32, 256)	10.9 M	25.9037	7.3324	22.9795	0.9989
SimVP (2, 6, 64, 256)	12.6 M	27.1322	7.5879	22.8999	0.9989
SimVP (2, 6, 32, 512)	28.3 M	25.8648	7.4242	22.9559	0.9990
SimVP (4, 8, 32, 256)	11.0 M	27.7265	7.4898	22.8378	0.9989
SimVP (2, 6, 64, 512)	32.2 M	25.3813	7.4156	22.9476	0.9990
SimVP (4, 6, 64, 512)	32.5 M	25.3018	7.4548	22.9364	0.9989
SimVP (4, 8, 64, 512)	43.1 M	25.8522	7.3213	23.0331	0.9990

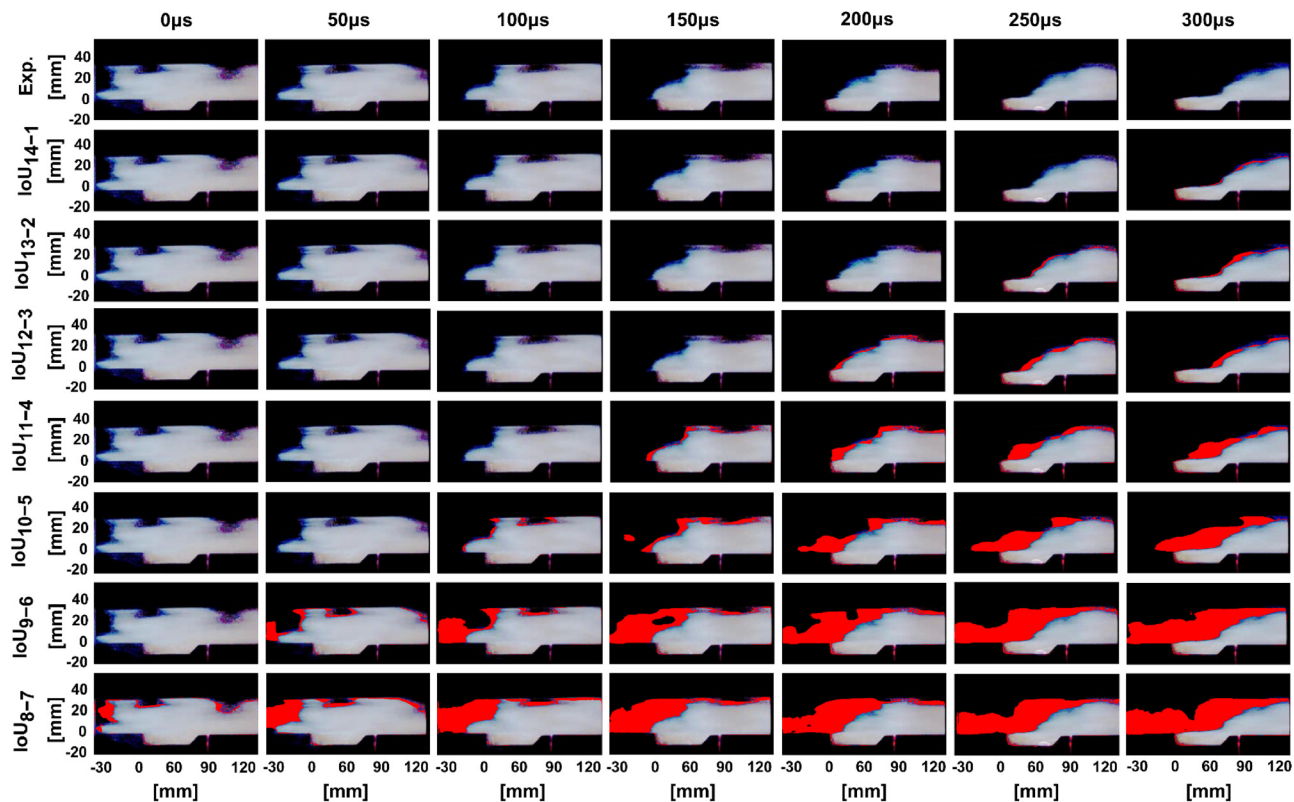


FIG. 5. Experimental kerosene-fueled scramjet combustion data and IoU imaging.

performance deteriorates as the prediction steps increases. From the case predicting five frames (IoU_{10-5}) to the case predicting 7 (IoU_{8-7}) frames, it can be observed that the flame reaction zone is incorrectly predicted within the cross-sectional isolator. Figure 7 quantifies this difference, showing a significant disparity in pixel intensity within the region from -50 to 120 mm. This indicates that areas that should have exhibited low-intensity combustion signals were incorrectly predicted as high-intensity signals. This occurs because, in the input images provided to the model, the flame reaction zone consistently appears within the cross-sectional isolator, making it difficult for the model to capture the transition of the reaction region. The flame experimental image at $150 \mu\text{s}$ contains crucial information, providing indications of the transition in the flame reaction zone, as the flame emission signal no longer appears within the cross-sectional isolator. It can be observed that once the model captures this transition (IoU_{11-4} , IoU_{12-3} , IoU_{13-2} , and IoU_{14-1}), the red regions are significantly reduced, indicating a substantial improvement in its prediction performance. The model can infer significant changes in the flame structure from a single image, demonstrating its strong capability in capturing spatiotemporal relationships.

Figure 7 presents the model's performance in predicting from 1 to 9 frames. Three indices were defined to quantify its accuracy, with their mathematical descriptions provided in Eqs. (2)–(6). After calculating 1791 image pairs, we plot the statistical data in the figure, indicating that the performance of our model declines as the number of prediction steps increases. Yet, it maintains a relatively satisfactory level of accuracy. PSNR for all cases is above 22.5, and according to

previous research,⁶³ it shows that the generated images have high prediction accuracy. However, when predicting 8 and 9 frames, the PSNR values are 22.5343 and 22.2013, respectively. Although the PSNR for 8-frame prediction remains slightly above 22.5, we do not recommend setting the prediction-to-input frame ratio greater than 1:1. This is because, when the ratio exceeds this threshold, the prediction accuracy is close to the minimum acceptable value ($\text{PSNR} = 22.5$). Additionally, the figure presents the IoU between the predicted and experimental images, facilitating the identification of regions that are accurately predicted and those that are not. According to some previous research,⁶⁷ an IoU of 0.65 and above suggests that the model has well reconstructed the images with high similarity. The averaged IoUs for 1791 sequences in all cases are above this threshold, indicating the model's high capability in predicting hypersonic combustion structures. When the prediction exceeds seven frames, the growth rates of MSE and MAE increase, indicating a greater degree of performance degradation.

Figure 8 demonstrates the flame structure tracking performance. In the combustion process of a scramjet engine, unintentional flame extinction is a critical issue. Predicting this phenomenon requires precise tracking of the flame structure. The SimVP model proposed in this paper demonstrates its potential in this regard. In the flameout scenario shown in Fig. 8, the reaction zone evolves with the high-speed flow of the fluid. The reaction zone of the flame underwent significant changes within $200 \mu\text{s}$, leading to flameout. The proposed model effectively captured this variation. As shown in Fig. 8, the encircled region

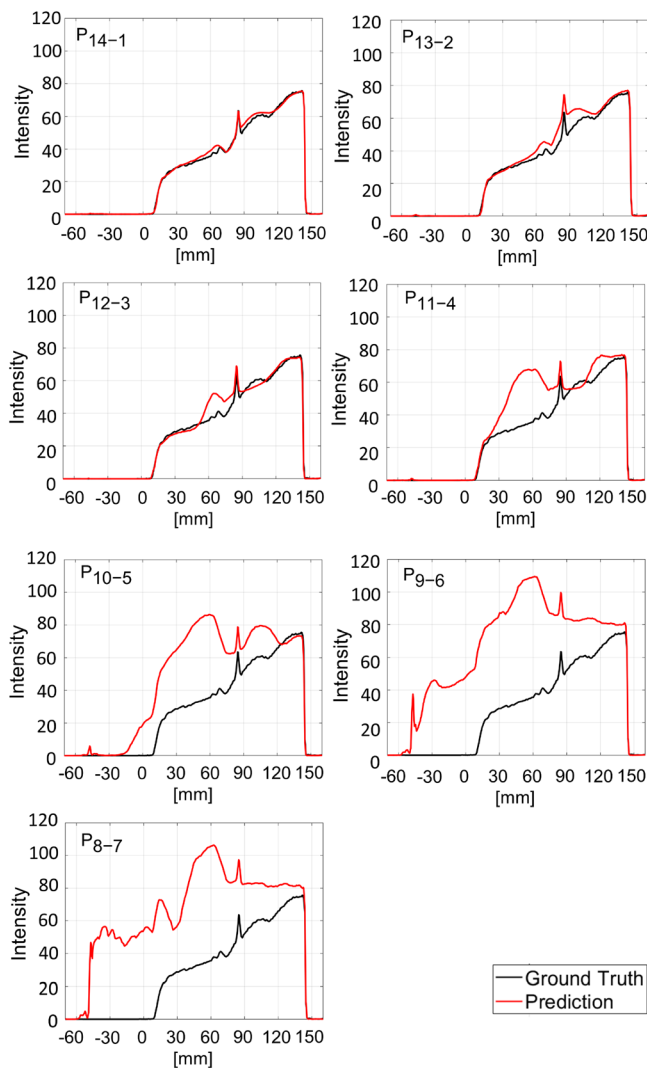


FIG. 6. Intensity profiles of the last experiment and predicted combustion images.

represents the morphological evolution of the flame extinction zone from the initial moment to the 200- μ s time point, with the intermediate images illustrating transitional stages of flame development. Temporospacial analysis reveals that the centroid of the extinction zone has undergone a displacement of 130 pixels, accompanied by significant morphological alterations, particularly an elongation along the cavity direction. Comparative analysis between experimental and predicted images demonstrates that these temporal and spatial variations have been successfully captured and predicted by the SimVP model.

Figure 9 presents the time-averaged scramjet combustion images and intermittency at a $m_k = 29.2$, $ER = 0.59$, b $m_k = 21.7$, $ER = 0.43$, and c $m_k = 12.2$, $ER = 0.24$. Each case includes (I) Time-averaged combustion images measured and predicted using SimVP; (II) the intermittencies of combustion image signals; (III) the profiles of time-averaged scramjet combustion (column-wise averaged), and (IV) those of intermittency (column-wise averaged). It was found that, according

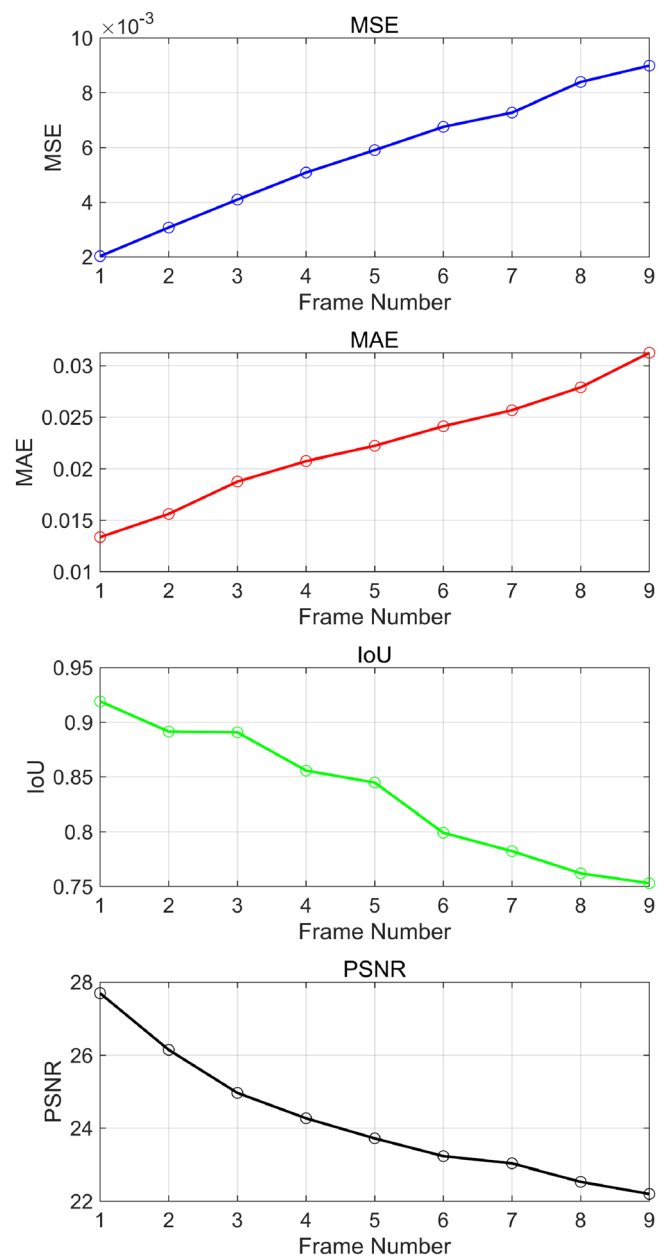


FIG. 7. Model performance based on 1791 test videos over frame number from 1 to 9.

to the column-wise averaged MAE results, the predictions agree with the experimental ones to within 1.12 for the mean and 0.01 for the intermittency in three conditions with different experimental parameters (m_k and ER).

C. Robustness test

In practical combustion scenarios, image quality may be severely affected because oil mist or smoke generated can prevent the camera

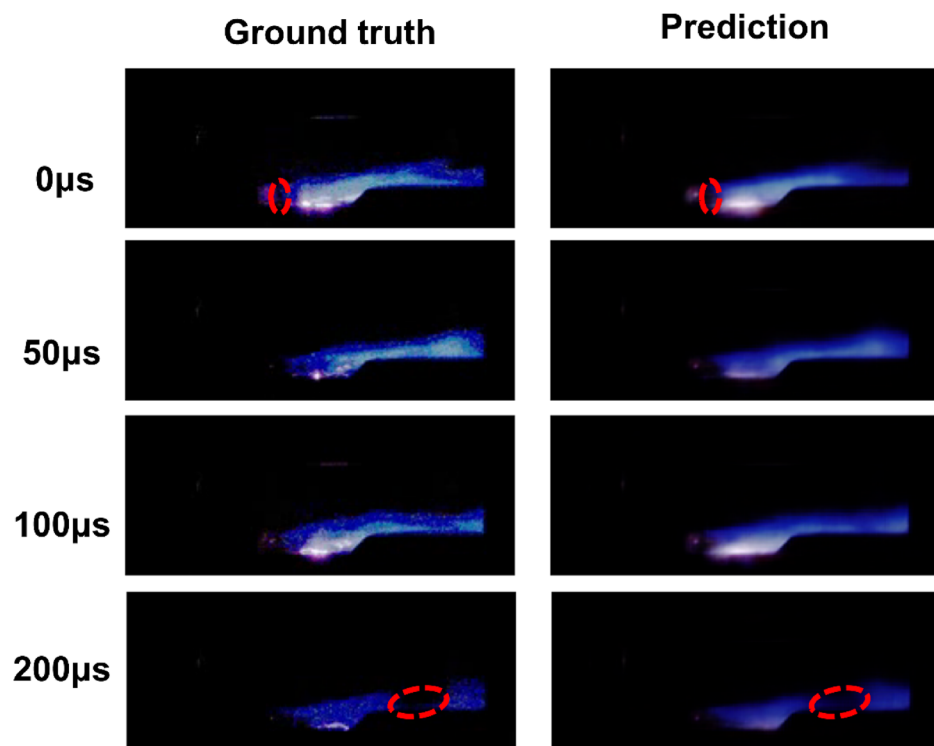


FIG. 8. The performance of flame structure tracking.

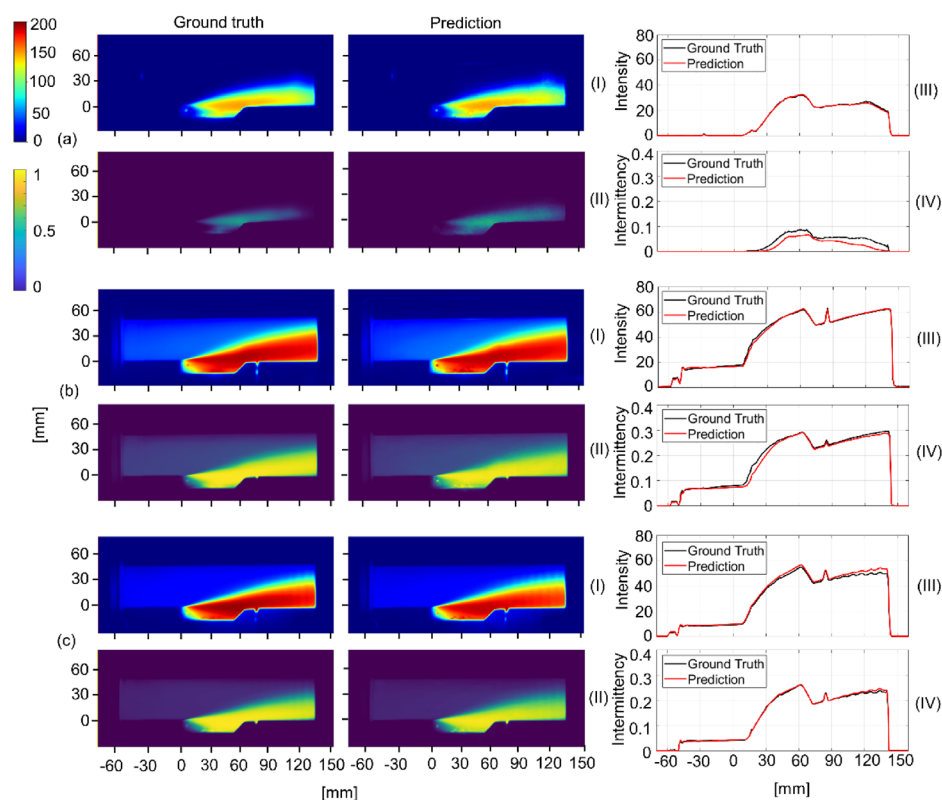


FIG. 9. The time-averaged scramjet combustion images and intermittency at a $m_k = 29.2$, $ER = 0.59$, b $m_k = 21.7$, $ER = 0.43$, and c $m_k = 12.2$, $ER = 0.24$. Each case includes (I) time-averaged combustion images measured and predicted using SimVP; (II) the intermittencies of combustion image signals; (III) the profiles of time-averaged scramjet combustion, and (IV) those of intermittency. Each case was measured in 597 test videos.

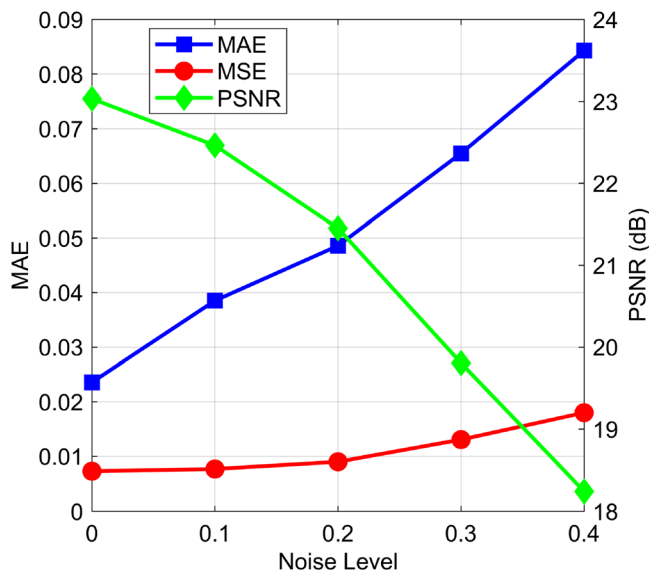


FIG. 10. Model performance under different noise levels in the test dataset.

from capturing clear combustion images.⁶⁸ The model should possess a certain level of data adaptability to maintain satisfactory performance even in harsh environments. Therefore, it is necessary to conduct robustness testing on the proposed model. By adding Gaussian noise to the test dataset, different levels of contaminated environments can be simulated, while the training data remains noise-free. The addition of noise can be expressed using the following formula:

$$\alpha' = \alpha + \sigma\tau, \quad (18)$$

where α' represents the noisy image and α represents the original image. σ is the noise level and τ is a normal random distribution

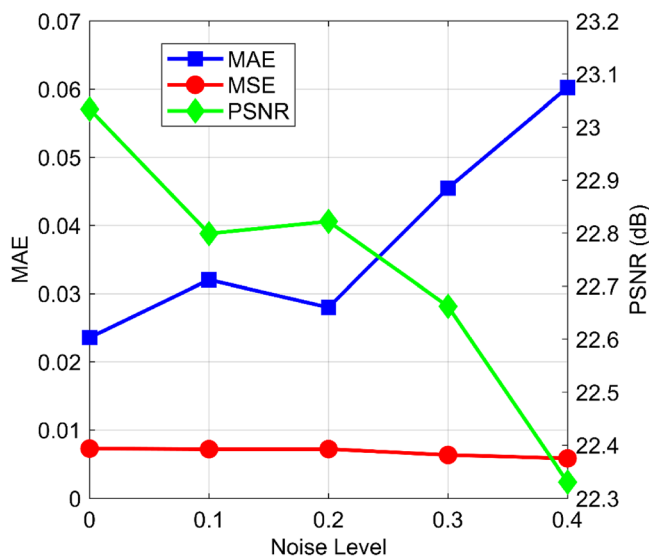


FIG. 11. Model performance under different noise levels in the train dataset.

within a range of -2.576 to 2.576 with 99% probability. The noise level σ in the training dataset is changed from 0 to 0.4 with a step size of 0.1. From the data in Fig. 10, it can be observed that PSNR exhibits moderate variations but still above 20 when the noise level is below 0.3. This indicates that the model can still achieve reliable predictions under conditions where the noise in the image is less than 30%. However, when the noise level exceeds 0.3, the model's performance deteriorates significantly. This is because the high proportion of noise makes the model prone to overfitting, mistakenly treating noise as useful information for prediction.

Figure 11 illustrates the model's performance when trained on datasets with varying levels of noise. The method used to add noise is the same as that applied to the test set. As the noise level increases, a decline in performance is observed. However, the degradation is less pronounced compared to scenarios where noise is added directly to the test set. This phenomenon can be observed by comparing the data in Figs. 10 and 11. For example, as the noise level in the dataset increases from 0% to 50%, the PSNR decreases by 5 dB. In contrast, the increase in noise in the training set results in a PSNR reduction of less than 1 dB. This indicates that the model possesses a certain degree of implicit denoising capability. Nevertheless, under high noise conditions—such as noise levels exceeding 20%—the model's performance deteriorates significantly.

In Fig. 12, we evaluated three different models; it is worth noting that PredRNN was excluded from the robustness test due to its failure to converge, and ConvLSTM was selected as a representative of RNN-based architectures. The results show that under the 7-frame prediction setting, when the noise level is below 20%, all three models exhibit similar performance, with SimVP performing the best. As the noise level increases, ConvLSTM experiences a significant drop in performance, whereas both TAU and SimVP demonstrate strong robustness, with TAU showing superior adaptability to high-noise environments.

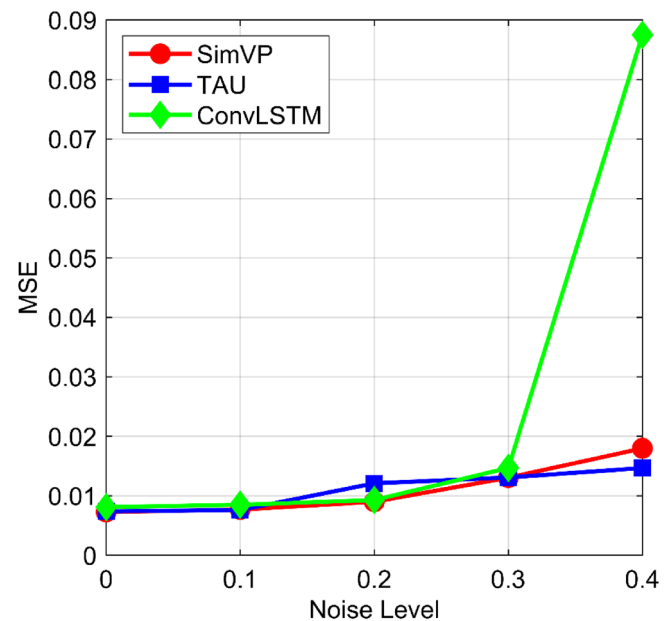


FIG. 12. Accuracy of the tested model under different noise levels in the test dataset.

TABLE V. Results of 7-frame prediction under different test flame conditions.

	Test 1	Test 2	Test 3
Train dataset flame conditions	(1)(2)	(1)(3)	(2)(3)
Test dataset flame condition	(3)	(2)	(1)
MSE ($\times 10^{-3}$)	1.224	14.352	7.196
MAE ($\times 10^{-3}$)	11.081	44.407	29.072
PSNR	29.121	20.446	22.980
SSIM ($\times 10^{-2}$)	99.988	99.790	99.896

D. Generalization capability test

To evaluate the generalization capability of the model, cross-condition testing was conducted using SimVP. The dataset employed in this study is summarized in Table I, consisting of three flame conditions: (1) $m_k = 29.2$, $ER = 0.59$; (2) $m_k = 21.7$, $ER = 0.43$; and (3) $m_k = 12.2$, $ER = 0.24$. Each condition contains a total of 2987 sequences. In the testing process, the training set comprises data from two of the three flame conditions, while the remaining condition is used for testing. Each condition is individually used as the test set to verify model performance, and the results are presented in the following table.

As shown in Table V, the best performance is achieved when flame conditions (1) and (2) are used as the training set, while the lowest accuracy is observed when conditions (1) and (3) are used for training. This discrepancy arises from the influence of data distribution in the training set on the test results. A significant mismatch between the distributions of the training and test sets can lead to degraded performance. In this study, the data distributions of conditions (1) and (2) are broader and more representative, resulting in better generalization. In contrast, the distributions of conditions (1) and (3) are more limited, leading to suboptimal performance.

V. CONCLUSION

This paper proposes a spatiotemporal flame prediction model based on SimVP to predict the flame structure of kerosene-fueled combustion collected from the scramjet cavity with the MCGA plasma experimental platform. We investigated the impact of four key structural parameters of the model—spatial block number (N_s), temporal block (N_t), hidden spatial state (h_s), and hidden temporal state (h_t)—on prediction performance and optimized the model architecture to achieve high-accuracy flame prediction. Additionally, it was found that the model has the capacity to generate flame sequences with different numbers of frames, even as the provided image information gradually decreases. Experiments demonstrate that the model can perform long-term prediction (7 frames) with limited information (8 frames) while maintaining a low MSE of 7.28×10^{-3} , MAE of 25.68×10^{-3} , and a high PSNR of 23.03 and SSIM of 0.99. Furthermore, the model was compared with other deep learning prediction models and the TAU model, and its efficiency and prediction performance were superior to all the tested models. In addition, the results of prediction within three different combustion cases ($m_k = 29.2$, $ER = 0.59$; $m_k = 21.7$, $ER = 0.43$, and $m_k = 12.2$, $ER = 0.24$) have been shown in this study. The statistical metrics show that under different operating conditions, the predictions agree with the experimental results

within 1.12 for the mean intensity and 0.01 for the intermittency. The model is also capable of achieving microsecond-level tracking of flame structures, with results showing that the extinction zone elongation along the cavity direction can be accurately captured. The robustness of the model was investigated by introducing noise into the test images. Based on the evaluation using three metrics, the model was able to tolerate up to a 30% noise level when predicting seven frames.

Future research will prioritize enhancing the model’s capability to predict abrupt flame structure transitions, with the objective of developing high-precision supersonic combustion flame mode prediction devices. Additionally, the model will be extended to diverse combustion scenarios to rigorously evaluate its generalization performance, supplemented by expanded experimental datasets to validate adaptability. Furthermore, given the intrinsic linkage between flame mode and chamber combustion behavior in supersonic flows, subsequent studies will integrate flame mode recognition with combustion dynamics prediction to establish a comprehensive framework for supersonic flame analysis.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qinglin Guan: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Xue Dong:** Formal analysis (lead); Funding acquisition (lead); Investigation (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (lead). **Liyang Sun:** Conceptualization (lead); Data curation (lead); Formal analysis (lead); Methodology (lead); Resources (lead); Software (lead). **Wei Zhang:** Conceptualization (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Software (lead); Supervision (equal); Writing – original draft (lead). **Jiajian Zhu:** Data curation (lead); Methodology (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (equal). **Tiangang Luo:** Conceptualization (equal); Data curation (equal); Project administration (equal); Supervision (equal); Writing – review & editing (equal). **Shuogui Xu:** Funding acquisition (equal); Project administration (equal); Supervision (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

APPENDIX: DESCRIPTIONS OF TESED MODELS

1. Convolutional LSTM network

The ConvLSTM network was reported by Shi *et al.*,⁴² and the main formulas of the ConvLSTM used in this paper are given as follows:

$$i_t = \sigma(W_{xi} * \chi_t + W_{hi} * h_{t-1} + W_{ci} \circ c_{t-1} + b_i), \quad (A1)$$

$$f_t = \sigma(W_{xf} * \chi_t + W_{hf} * h_{t-1} + W_{cf} \circ c_{t-1} + b_f), \quad (A2)$$

$$c_t = f_t \circ c_{t-1} + i_t \circ \tanh(W_{xc} * \chi_t + W_{hc} * h_{t-1} + b_c), \quad (A3)$$

$$o_t = \sigma(W_{xo} * \chi_t + W_{ho} * h_{t-1} + W_{co} \circ c_t + b_o), \quad (A4)$$

$$h_t = o_t \circ \tanh(c_t), \quad (A5)$$

where $*$ represents the convolution operation and $\circ$ is the Hadamard product. In the context of the model, i_t , f_t , and o_t correspond to the input gate, forget gate, and output gate, respectively. These gates regulate the incorporation of new information, the discarding of irrelevant information, and the final output of a single ConvLSTM cell. It is noteworthy that in the context of ConvLSTM networks, all three gates are represented as three-dimensional tensors representing images with a shape of (C, H, W), where C, H, and W denote the number of channels, height, and width, respectively. X_t , h_t and c_t are the input, hidden layer memory state, and memory cell output at timestamp t in the ConvLSTM framework, the memory cell output c_t serves as a dynamic accumulator of state

information, which is selectively updated and controlled through a series of self-parameterized gates. When a new input is introduced, its information is accumulated into the memory cell if the input gate i_t is activated. Concurrently, the previous cell state c_{t-1} can be selectively “forgotten” if the forget gate is activated. The propagation of the updated cell state to the final hidden state h_t is further regulated by the output gate. The ConvLSTM encoding-forecasting architecture for scramjet combustion is as shown in Fig. 13. As illustrated in the figure, the Conv-LSTM encoder compresses the combustion image sequence into a hidden state tensor, which is then expanded by the ConvLSTM forecaster to generate the final prediction. This architecture resembles the LSTM future predictor model proposed by Srivastava *et al.*,⁶⁹ with the key difference that both the input and output elements in our model are 3 D tensors, preserving complete spatial information. Due to the presence of multiple stacked ConvLSTM layers, the network is capable of predicting sequences in complex dynamic systems, similar to its application in solving the precipitation nowcasting problem by Shi *et al.*⁴²

2. PredRNN network

PredRNN was proposed by Wang *et al.*⁴³ and it adopted ConvLSTM by combining standard temporal memory c_t^l and spatio-temporal memory M_t^l in the layer l at timestamp t . The equations of this network are shown as follows:

$$i_t = \sigma(W_{xi} * \chi_t + W_{hi} * h_{t-1} + W_{ci} \circ c_{t-1} + b_i), \quad (A6)$$

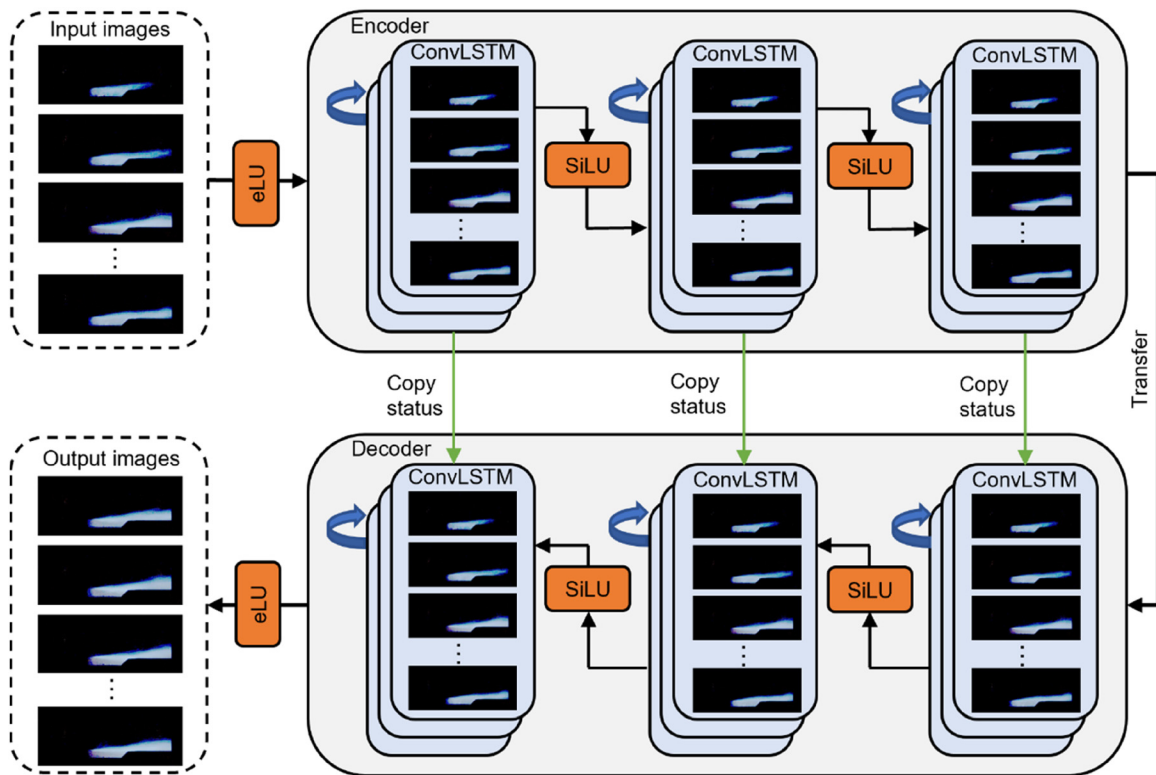


FIG. 13. The encoding-forecasting architecture tested in the ConvLSTM model.

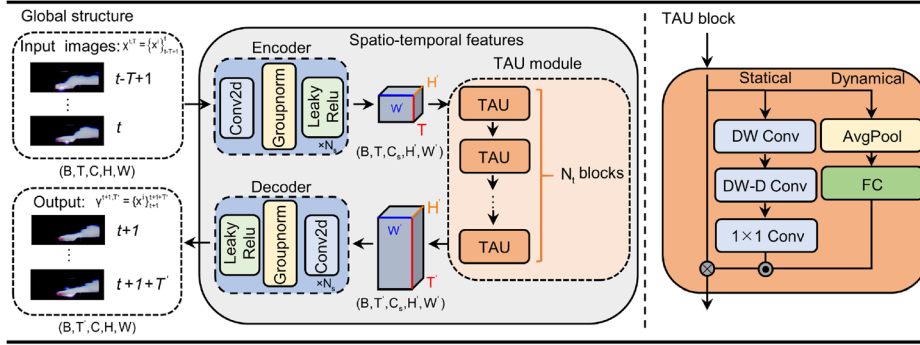


FIG. 14. The overall framework of the TAU network.

$$f_t = \sigma(W_{xf} * \chi_t + W_{hf} * h_{t-1} + W_{cf} * c_{t-1} + b_f), \quad (A7)$$

$$c_t^l = f_t \circ c_{t-1}^l + i_t \circ \tanh(W_{xg} * \chi_t + W_{hg} * h_{t-1}^l + b_c), \quad (A8)$$

$$i_t^l = \sigma(W_{xi}' * \chi_t + W_{mi}' * M_{t-1}^{l-1} + b_i'), \quad (A9)$$

$$f_t' = \sigma(W_{xf}' * \chi_t + W_{mf}' * M_{t-1}^{l-1} + b_f'), \quad (A10)$$

$$M_t^l = f_t' \circ M_{t-1}^{l-1} + i_t^l \circ \tanh(W_{xi}' * \chi_t + W_{mi}' * M_{t-1}^{l-1} + b_c'), \quad (A11)$$

$$o_t = \sigma(W_{xo} * \chi_t + W_{ho} * h_{t-1}^l + W_{co} * c_t^l + W_{mo} * M_t^l + b_o), \quad (A12)$$

$$h_t^l = o_t \circ \tanh(W_{1 \times 1} * [c_t^l, M_t^l]). \quad (A13)$$

In PredRNN, the memory state was proposed to maintain representations from the raw input sequence and different-level convolutional layers. Each layer has distinct input-to-state and state-to-state convolutional parameters, while maintaining a spatio-temporal memory cell that is updated separately and iteratively as information propagates through the current node. Compared with ConvLSTM, the input gate, forget gate, and output gate no longer depend on the hidden states and cell states from the previous time step at the same layer, and they rely on the hidden state h_t^l and memory state M_t^l . The fused spatiotemporal memory is used as the final hidden state in Eq (A12). The fused spatiotemporal memory is obtained by applying a 1×1 convolution layer for dimension reduction to M_t^l and c_t^l .

3. Temporal Attention Unit

Temporal Attention Unit was reported by Tan *et al.*⁵¹ and this framework is a variant of SimVP shown in Fig. 14. This network shares the same spatial encoder and decoder as SimVP but employs the TAU block to capture temporal correlations within the image sequence. As shown in Fig. 14, the network architecture replaces SimVP's inception module with a sequence of N_t TAU blocks. Each TAU block incorporates intra-frame static attention and inter-frame dynamic attention. Small kernel depth-wise convolutions (DW Conv), depth-wise convolutions with dilations (DW-D Conv), and 1×1 convolutions were employed to get statistical attention. An AvgPool layer and a fully connected layer in a single TAU block are applied to get dynamic attention. The operation in TAU blocks can be expressed as follows:

$$z_k = \text{TAU}(z_{k-1}), \quad N_s \leq k \leq N_s + N_t, \quad (A14)$$

$$\text{SA} = \text{PW Conv}(\text{DW} - \text{D Conv}(\text{DW Conv}(H))), \quad (A15)$$

$$\text{DA} = \text{FC}(\text{AvgPool}(H)), \quad (A16)$$

$$H' = (\text{SA} \otimes \text{DA}) \odot H. \quad (A17)$$

where $H \in \mathbb{R}^{B \times (T' \times C) \times H \times W}$ is the hidden feature that will be fed into the TAU block, $\text{SA} \in \mathbb{R}^{B \times (T' \times C) \times H \times W}$, $\text{DA} \in \mathbb{R}^{B \times (T' \times C) \times 1 \times 1}$ denote the static and dynamic attention, FC and AvgPool are fully connected layers and the average pooling. The Kronecker product and the Hadamard product are represented by $\otimes$ and $\odot$, respectively.

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